

BLY 111: GENERAL BIOLOGY I

(3 CREDITS UNIT)

BY

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Course Content

General tenets of plant life, general tenets of animal life, cells structure and organization, diversity and characteristics of living things, general reproduction and interrelationship of organisms, heredity and evolution, elements of ecology, habitats and diversity.

Objectives

At the end of these lectures, students should be able to;

- Know the general tenets of animal life
- Understand the different characteristics of living things
- Know how to classify living things
- Know how to classify invertebrates and vertebrates
- Know the characteristics features of invertebrates and vertebrates
- Know what a chordate means

GENERAL TENETS OF ANIMAL LIFE & DIVERSITY OF LIVING THINGS

The most outstanding general features/principles in life's history include;

Chemical uniqueness

-Complexity and hierarchical organization

-Reproduction (heredity and variation)

Possession of a genetic program

-Metabolism

-Development

-Environmental interaction

-Movement

CHEMICAL UNIQUENESS

Living systems demonstrate a unique and complex molecular organization. Living systems assemble large molecules known as macromolecules that are far more complex than the small molecules of non-living matters.

These macromolecules are composed of the same kinds of atoms and chemical bonds that occur in non-living matter and obey all fundamental laws of chemistry. It is only the complex organizational structure of these macromolecules that makes them unique. There are 4 major categories of biological macromolecules, nucleic acids, protein, carbohydrate, and lipid.

These macromolecules differs in the structures of their component parts, the kinds of their component parts, the kind of chemical bonds that link their subunits together and their functions in the living system.

- For example, proteins contain about 20 specific kinds of amino acid

COMPLEXITY AND HIERARCHICAL ORGANIZATION

Living system demonstrate a unique and complex hierarchical organization. Non-living matter is organized at least into atoms and molecules and often has a higher degree of organization as well. However, atoms and molecules are combined into patterns in the living world that do not exist in the non-living world.

-In the living world, there are hierarchy of levels that includes, in ascending order of complexity, macromolecules, cells, organisms, populations and species.

Sub units linked together by peptide bond in a linear sequence.

The nucleic acids, carbohydrates and lipids likewise contain characteristic bonds that link variable subunit. This organization gives living systems both a biochemical unity and great potential diversity. Linnaeus's scheme of arranging organisms into an ascending series of groups of ever-increasing inclusiveness is a **hierarchical system of classification**. Major taxa into which organisms are grouped were given one of several standard taxonomic ranks to indicate the general degree of inclusiveness of the group. (Kingdom, Phylum, Subphylum, Class, Subclass, Order, Suborder, Family, Subfamily, Genus, Species)

Linnaeus's system for naming species is known as binomial nomenclature. Each species has a Latinized name composed of two words (hence binomial) printed in italics (or underlined if handwritten or typed). The first word names the genus, which is capitalized, the second word is the species (e.g. Volvox globatoris or *Volvox globatoris*)

Volvox globatoris a multicellular chlorophyton that illustrate three different levels of the biological hierarchy, cellular, organismal and populational. Each individual organism contains cells embedding in a gelatinous matrix.

-Each levels builds on the level below it and has its own individual structure, which is also often hierarchical. Within the cell, for example, macromolecules are compounded into structures such as ribosomes, chromosome and membranes, and these are likewise combined in various ways to form even more complex subcellular structures called organelles- such as mitochondria. The organismal level also has a hierarchical sub-structure; cells combine to form organs, which likewise combine to form organ systems.

Cells are the smallest units of the biological hierarchy that are semi-autonomous in their ability to conduct basic functions including reproduction. Replication of molecules and subcellular components occur only within cellular context not independently.

Cells are therefore considered the basic units of living system. Cells from an organism can be isolated and caused to grow and multiply under laboratory conditions in presence of nutrients alone. The semi-autonomous replication is not possible for any individual molecules or subcellular components, which require additional cellular constituent for their reproduction.

REPRODUCTION

Living systems can reproduce themselves. Life does not arise spontaneously but comes only from prior life, through reproduction. Although life certainly originated from nonliving matter, at least at each level of the biological hierarchy, living forms reproduce to generate others like themselves. Genes are replicated to produce new genes.

Cells divide to produce new cells. Organisms reproduce sexually to produce new organisms. Populations may become fragmented to produce new populations, and species may split to produce species through a process called speciation. Reproduction at any hierarchical level usually features an increase in numbers.

Generally, two methods of reproduction is shown in animals: asexual and sexual reproduction. Asexual reproduction involves binary fission, multiple fission, fragmentation, colony formation, etc while sexual reproduction involves the formation of male and female gametes from the male and female sexes. Sometimes, the two sexes are formed within one individual and such individuals are termed **hermaphrodites**

Individual genes, cells, organisms, populations or species may fail to reproduce, but reproduction is nonetheless an expected property of these individuals.

Reproductions at each of these levels show the phenomena of heredity and variation. Heredity is the faithful transmission of traits from parents to offspring, usually observed at organismal level.

Variation is the production of differences among the traits of different individuals. In a reproductive process, properties of descendants resemble those of their parents to varying degrees but usually are not identical to them.

POSSESSION OF A GENETIC PROGRAM

A genetic program provides exact inheritance. Structures of the protein molecules needed for organismal development and functioning are encoded in nucleic acid. For animals and most other organisms, genetic information is contained in deoxyribonucleic acid (DNA). DNA is a very long linear chain of subunit called nucleotides, each of which contains a sugar phosphate (deoxyribose phosphate) and one of four nitrogenous bases (adenine, cytosine, guanine or thymine abbreviated A C G T respectively).

METABOLISM

Living organism maintain themselves by acquiring nutrients from their environment. The nutrients are used to obtain chemical energy and molecular components for building and maintaining the living system. These essential chemical processes is called metabolism. They include digestion, acquisition of energy (respiration), and synthesis of molecules and structures. It involves both anabolic and catabolic reactions. These reactions include synthesis of carbohydrates, lipids, nucleic acids and proteins and their constituent parts and cleavage of chemical bonds to recover energy stored in them.

DEVELOPMENT

All organisms pass through a characteristic life cycle. Development describes the characteristic changes that an organism undergoes from its origin (usually fertilization of egg by sperm) to its final adult form.

Development usually features changes in size and shape, and differentiation of structures within an organism. Even the simplest one-celled organism e.g. Amoeba grow in size and replicate their component parts until they divide into two or more cells. Multicellular organisms undergo more dramatic changes during their lives.

Different developmental stages of some multicellular forms are so dissimilar that they are hardly recognizable as belonging to the same species.

Embryos are distinctly different from juvenile and adult forms into which they develop. Even post embryonic development of some organisms include stages dramatically different from each other. The transformation that occurs from one stage to another is called Metamorphosis. There is little resemblance for example among the egg, larval, pupal and adult stages of metamorphic insects. For example, in the life cycle of butterfly, the egg, larva also called caterpillar are completely different; likewise the pupa and adult.

ENVIRONMENTAL INTERACTION

All animals interact with their environments. The study of organismal interaction with an environment is called ecology. Certain environmental factors influence the geographical distribution and abundance of animals. For example, Toads and Frogs emerge usually at the onset of rains in stagnant pool of water. Ecology also reveals how an organism perceives environmental stimuli and responds in appropriate ways by adjusting its metabolism and physiology.

Toads and frogs go into a period of hibernation when rainfall ceases. All organisms respond to environmental stimuli, a property called irritability.

The stimulus and response may be simple for e.g. Amoeba moving from or towards a light source or it may be quite complex such a bird responding to a complicated series of signals in a mating ritual.

MOVEMENT

Living systems and their parts show precise and controlled movements arising from within the system. The energy that living systems extract from their environments permit them to initiate controlled movements. Such movements at the cellular level are essential for reproduction, growth and many responses to stimuli in all living forms and for development in multicellular ones.

On a larger scale, entire populations or species may disperse from one geographic location to another one over time through their powers of movement.

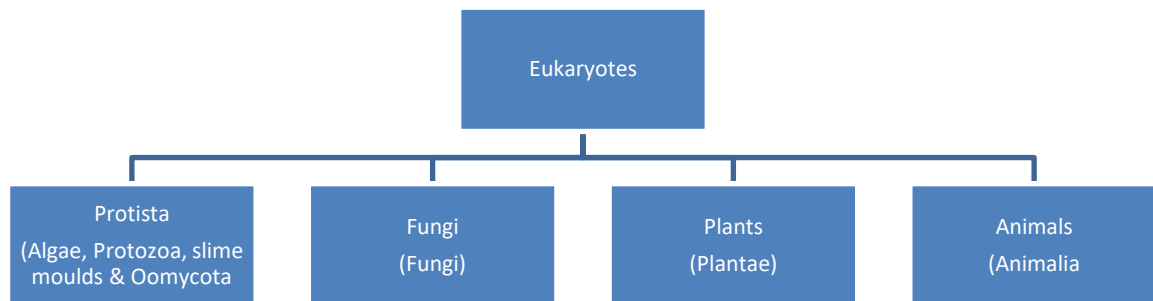
DIVERSITY OF LIVING THINGS

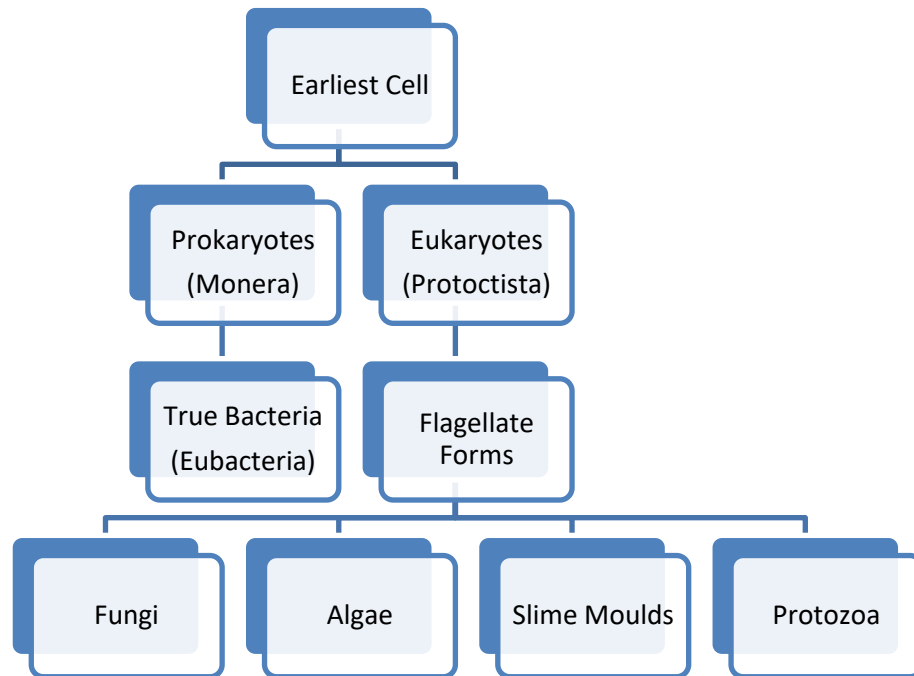
All cellular organisms were naturally divided into two major groups: The **Prokaryotes** and **Eukaryotes**. **Prokaryotes** appeared about 3500 million years ago and comprise a variety of organisms collectively known as bacteria. **Eukaryotes** appeared about 2000 million years ago and probably evolved from Prokaryotes and comprise Protista, fungi, green plants and animals.

The cells of Prokaryotes, their genetic material (DNA) is not enclosed by nuclear membranes and lies free in the cytoplasm. The cells of eukaryotes (eu – true) are much more complex and are characterized by a true nucleus, that is genetic material enclosed by membranes (the nuclear envelope) to form a definite, easily recognizable structure. Many more fundamental differences exist between Prokaryotes and Eukaryotes as shown below. Prokaryotes (pro – before, karyon – nucleus) lack true nuclei.

FEATURES	PROKARYOTES	EUKARYOTES
Cell Size	Average diameter 0.5 – 5µm	Up to 40µm diameter
Form	Unicellular or filamentous	Unicellular, filamentous or truly multicellular
Genetic Material	Circular DNA lying naked in the cytoplasm, No true nucleus or chromosomes. No nucleolus	Linear DNA associated with proteins and RNA to form chromosomes in the nucleus. Nucleolus in nucleus
Protein Synthesis	70S Ribosomes (smaller), No endoplasmic reticulum involved	80S Ribosomes (larger). Ribosomes may be attached to endoplasmic reticulum
Organelles	Few organelles. None are surrounded by an envelope	Many organelles. Envelope-bound organelles present e.g. nucleus, mitochondria, chloroplasts
Cell Walls	Rigid and contains polysaccharides with amino acid. Murein is main strengthening compound.	Cell walls of green plants and fungi rigid and contain polysaccharides. Cellulose is main strengthening compound of plant walls, chitins of fungal walls.
Flagella	Simple lacking microtubules. Extracellular (not enclosed by cell surface membrane). 20nm diameter	Complex, with 9+2 arrangement of microtubules. Intracellular (surrounded by cell surface membrane). 200nm diameter

Respiration	Mesosomes in bacteria, Except Cytoplasmic membranes in blue-green algae	Mitochondria for aerobic respiration
Photosynthesis	No Chloroplasts. Takes place on membranes which show no stacking	Chloroplasts containing membranes which are usually stacked into lamellae or grana
Nitrogen Fixation	Some have the ability	None have the ability





KINGDOM - PROTOCTISTA

(Greek word: protos – very first, ktistos – to establish)

Protoctists include all eukaryotic organisms which are no longer classified as animals, plants or fungi. Many are unicellular organisms or collection of similar cells. The kingdom includes all algae, protozoa, slime moulds and the oomycota. Algae and protozoa are adapted for aquatic habitats.

	Rhizopoda (rhizopods)
	Zoomastigina (flagellates)
	Apicomplexa (sporozoans)
	Ciliophora (ciliates)
	Oomycota (oomycetes)
	Euglenoidea (euglenoids)
Protocista (Kingdom)	Chlorophyta (green algae)
	Rhodophyta (red algae)
	Bacillariophyta (diatoms)
	Diriophyta (dinoflagellates)
	Xanthophyta (yellow green algae)

ALGAE

The algae form a large group of protocistans of great biological importance and significance to humans. Their bodies lack true stems, roots and leaves. Originally, they were classified with fungi in the Division **Thallophyta**. Algae are the best oxygen-producing photosynthetic eukaryotes. Algae are **autotrophic plants** i.e. they manufacture their own food with the help of chlorophyll. The body of an algae is composed of a true parenchymatous tissue and its cell wall is made up of true cellulose. Algae are found in water, or wet substrata and its reserve carbohydrate consists of starch. Examples of Algae include *Chlamydomonas*, *Volvox*, *Spirogyra*, etc

PROTOZOA

The protozoa (protos – first; zoon – animal) are a diverse group of organisms. They are found in all environments where water is present.

- Protozoa are unicellular in nature with much simple structure.
- Most protozoa possess a definite shape except phylum – Rhizopoda e.g *Amoeba* whose shape is variable.

- Adult protozoa move using various structures depending on the species – flagella, pseudopodia or cilia..
- They possess one nucleus enclosed in a nuclear membrane except phylum Ciliophora where *Paramecium* has two nuclei (macronucleus and micronucleus).
- Asexual Reproduction in protozoa occur by binary fission. Sexual reproduction only occur in phylum Ciliophora by conjugation (e.g *Paramecium*).
- Examples of Protozoa include *Amoeba*, *Euglena*, *Paramecium*, *Trypanosoma*, *Plasmodium*, etc.

KINGDOM – FUNGI

The fungi are a large and successful group of organisms of about 80,000 named species. Fungi are of major importance for the essential role that they play in the biosphere, and for the way in which they have been exploited by humans for economic and medical purposes. Fungi include numerous moulds growing in damp organic matter (such as bread, leather, decaying vegetation and dead fish), unicellular yeast which are abundant on sugary surfaces of ripe fruits and many parasites of plants. The study of fungi is called **Mycology**.

- Fungi are eukaryotes that lack chlorophyll and are therefore heterotrophic like animals
- They have rigid cell walls containing chitin and are non-motile like plants
- Traditionally, they have been regarded as plants but more modern classification places them in a separate kingdom.
- Fungi consist of fine, tubular branching threads called **hyphae**, the whole mass being called a **mycelium**.
- If carbohydrate is stored, it is usually as glycogen, not starch
- Asexual Reproduction is by mean of conidia or sporangia containing spores
- Examples of fungi include *Rhizopus stolonifer* (common bread mould), *Mucor*, *Penicillium*, *Saccharomyces* (yeast).

• KINGDOM – PLANTAE

- Plants are described as autotrophic eukaryotes different from fungi which are heterotrophic. The only other autotrophic eukaryotes, apart from plants are algae
- Bryophyta (Mosses and Liverwort)
- Pteridophyta (Ferns and Allies)

- PLANTAE Gymnospermophyta (Conifers)
- (Kingdom) Angiospermophyta (Flowering Plants)

PHYLUM – BRYOPHYTA

Bryophytes which were the first plants to colonise the land were generally thought to have evolved from green algae. The Bryophytes are relatively poorly adapted to life on land, so are mainly confined to damp, shady places.

- They lack vascular tissues, that is no xylem or phloem
- Their body is thallus or differentiated into simple leaves and stems.
- No true roots, stems or leaves are found, The gametophyte is anchored by filamentous rhizoids. There is alternation of generations in which the gametophyte generation is dominant.
- The sporophyte is attached to and is dependent upon the gametophyte for its nutrition.
- Spores are produced by the sporophyte in a spore capsule on the end of a slender stalk above the gametophyte.
- Examples of bryophytes are Mosses and Liverworts

PHYLUM - PTERIDOPHYTA

Pteridophytes commonly comprise ferns, clubmosses and horsetails. They are found usually in damp, shady habitats. Some are found growing in full sunlight, wet rocks and pastures near streams and ponds.. These are the earliest known vascular plants containing vascular tissues. Vascular tissues is a feature of sporophyte generation. This generation is small in the bryophytes and is dependent on the gametophyte.

- Pteridophyte possess alternation of generation in which the sporophyte generation is dominant.
- Gametophyte is reduced to a small, simple prothallus
- Sporophyte has true roots, stems and leaves with vascular tissues

PHYLUM – GYMNOSPERMOPHYTA/ CONIFEROPHYTA

- These are one of the seed bearing plants.
- Gymnosperms bear naked seeds meaning that the seeds are exposed (not enclosed in an ovary).
- In Gymnosperms, ovules which later develop into seeds are located on the surfaces of specialized leaves called **megasporophylls** or **ovuliferous scales**.
- These are arranged in cones on which the sporangia and spore develop.
- In Gymnosperms, no fruit is formed because there is no ovary.
- The Xylem tissue contains only tracheids with no vessels while there is no companion cell in the Phloem. Only albuminous cells similar in function to companion cells are found but of different origin.
- Examples of Gymnosperms are Conifers, Cycads, Ginkgos, etc

PHYLUM – ANGIOSPERMOPHYTA (Flowering Plants)

- These are seed bearing plants whose seeds are enclosed in an ovary thus giving more protection to the gametophyte and subsequent zygote
- They are commonly known as flowering plants because they produce flowers in which the sporangia and spores develop. That is the produce flower instead of cones as is the case with the Gymnosperms. This has enable many of them to utilize insects, and occasionally birds or even bats as a agents of pollination
- After fertilization, the ovary develop into a fruit in Angiosperms.
- Xylem contains vessels while Phloem contains companion cells which make them quite different from Gymnosperms.

- Angiosperms are better adapted to life on land than any other plants: could be found in very dry habitats depending on the species.
- The two classes of Angiosperms are **Dicotyledoneae** and **Monocotyledoneae**.

KINGDOM – ANIMALIA

- All animal phyla are composed of multicellular, heterotrophic organisms. Only the sponges (Phylum – Porifera) do not form true tissues but all other animals within the multicellular body similar cells operate collectively and become specialized functionally to form tissues. Many different tissues can be formed, each performing different functions. This is called differentiation or division of Labour within the organism. Hence the tissues perform specific tasks more effectively than individual cells.
- A number of tissues may work together as an **organ**, and a group of organs working together forms **organ-system**. Just as cells are unable to act independently within a tissue, so organs and organ systems are subordinate to some means of coordination by the body Porifera (Sponges)

	Coelenterata (Hydra)
	Platyhelminthes (Tapeworm)
ANIMALIA	Nematoda (Roundworm)
(Kingdom)	Annelida (Earthworm)
	Arthropoda (insects)
	Echinodermata (Starfish)
	Chordata (Mammals)

PHYLUM - PORIFERA

Characteristic Features

Some cellular differentiation occur but there is no true tissue organization as in other animals.

The body wall of sponges possess two layers of cells: outer pinacoderm and inner choanoderm (of flagellate collar cells).

Numerous pores (called Ostia) are found in the body wall

All adults are sessile and attached to substratum

Mostly marine even though all species are aquatic

The skeletal structures are calcareous or siliceous or horny fibres.

They possess radial symmetry or none Asexual reproduction occurs by budding while sexual reproduction occurs by production of eggs and sperm. They are all hermaphrodites, even though most are protandrous

There is no organ or true tissue, digestion is intracellular, excretion and respiration occur by diffusion

There is no differentiated nervous system

Examples of sponges include Leucosolenia, Sycon PHYLUM - CNIDARIA (Coelenterates)

Characteristic Features

Coelenterates are diploblastic animals whose body wall is composed of two layers of cells an outer ectoderm and an inner endoderm; these two layers are separated by a structure less, gelatinous layer of mesoglea.

It is in this group that tissue level of organization is achieved

They possess a single cavity corresponding to the coelenterons; primarily inhalant, secondary exhalent.

Coelenterates possess a single opening for ingestion and egestion.

They exhibit radial symmetry.

Coelenterates possess two (2) forms: the sedimentary **polyp forms** which may be solitary colonial **medusoid forms**, free swimming and solitary. Nervous system consists of a collection of cells forming an irregular net or plexus.

Asexual reproduction occurs by budding or strobilation while sexual reproduction produces a charactersistics **planula larva**.

Polymorphism is exhibited, but most individuals are reduced to either the hydroid (e.g Hydra) or medusoid type {e.g Aurelia (jellyfish)}

PHYLUM –PLATYHELMINTHES

GENERAL CHARACTERISTICS

- The Adult body is three-layered (Triploblastic).
- They exhibit bilateral symmetry with a definite polarity of the anterior and posterior ends.
- Their bodies are flattened dorso ventrally, hence the name flatworm.
- They are acoelomate animals (without body cavity).
- Central nervous system is anteriorly placed with very simple network and ganglia.
- Excretory system has branching tubes ending in flame cells. Asexual reproduction occur by fragmentation, Sexually, they have complex hermaphroditic reproductive system (monoecious).
- Digestive system is incomplete with mouth without anus in some (eg *Planaria*), may be branched in some(eg *Fasciola*) and absent in cestodes (tapeworm)
- Examples of members of this phylum are ***Planaria***, ***Fasciola*** (liver fluke), ***Taenia*** (tapeworm)

PHYLUM- NEMATODA

GENERAL CHARACTERISTICS

- Nematodes are triploblastic, pseudocoelomic animals.
- They exhibits bilateral symmetry and are unsegmented in nature.
- They are rounded in transverse section, hence the name round worm and are pointed at both end.
- Sexes are separate (dioecious).
- Digestive system is complete with mouth and anus Excretory system consists of a pair of lateral excretory canals ending in renette cells
- Nematodes are free living while many are important parasites of plants and animals.
- Examples of nematodes include *Ascaris* (roundworm), *Trichuris* (whipworm), *Dracunculus* (Guineaworm), etc

PHYLUM – ANNELIDA

CHARACTERISTIC FEATURES

- Annelids are triploblastic, coelomate animals (with true body cavity)
- They are bilaterally symmetrical and metamerically segmented animals
- Excretory organs are segmental, ectodermal in origin, ciliated and are called Nephridia
- Annelids possess chaetae made up of chitin arranged segmentally (except Leeches).
- The central nervous system is made up of paired supra-oesophageal ganglia connected to the double ventral nerve cord by commissures.
- They possess a definite cuticle secreted by the ectoderm
- Some species are dioecious (eg *Nereis*) while others are hermaphroditic eg *Lumbricus* (earthworm), ***Hirudos*** (Leeches)

PHYLUM - MOLLUSCA

CHARACTERISTIC FEATURES

- Molluscs are unsegmented, triploblastic coelomate animals
- They are usually bilaterally symmetrical.
- Molluscan body is divided into a head, ventral muscular foot and dorsal visceral hump (mass).
- Respiratory pigment is usually haemocyanin.
- Circulatory system has a heart with an open haemocoelic system
- Nervous system consists of circumoesophageal ring with cerebral and pleural ganglia, pedal cords and visceral loop.
- The skin of mollusk is soft, and over the hump it forms a mantle which secretes a calcareous shell.
- They are basically oviparous and produce trochophore larva
- Examples of Molluscs include Land snails, Oyster, Clams, Squids, etc

Phylum : Arthropoda

The phylum arthropoda (Gr. Arthros, jointed + podos, foot) contains over 90 per cent of the known species of animals, or approximately 950,000 species. It includes the crabs, shrimps, barnacles and other crustaceans (class **Crustacea**), the insects (Class **Insecta**); the spiders, scorpions, ticks, etc. **Class Arachnida**, the centipedes (sub-class **Chilopoda**, the millipedes (subclass **Diplopoda**), and some less familiar and fossil forms.

The body is segmented externally in varying degree, and the appendages are jointed; both are variously specialized in form and function in different kinds of arthropods in keeping with their manner of life. All external surfaces are covered with an organic exoskeleton. The nervous system, eyes and other sense organs are usually well developed, providing quick responses to stimuli. This is the only invertebrate phylum having many members able to live on land, away from moist surroundings; and the insects are the only invertebrates able to fly.

Arthropods of different kinds inhabit every sort of environment and some species are enormously abundant. Various they occupy the air and land, the soil, and fresh, salt, or brackish waters. Some are parasites of plants, and others live on or in many animals

Crabs, lobsters, and shrimps serve as human food, smaller crustaceans are the basis of aquatic food chains, and insects and spiders are eaten by land vertebrates. Insects are human beings' chief competitors for crops, stored foods, clothes, and other property, although many kinds are beneficial. Certain insects, ticks, and mites carry diseases to humans, their domestic animals, and their crops.

Characteristic Features

- Arthropods show bilateral symmetry with body segmented and grouped into head, thorax, and abdomen (distinct or fused).
- One pair of appendages (or none) occur per segment, each with few or many hinge joints and opposed muscles; commonly specialized, sometimes reduced, rarely lacking.
- Arthropod exoskeleton contains chitin, often hardened and molted at intervals.
- They possess many and varied muscle fibres mainly striated fibers,
- Arthropods possess complete digestive tract with chewing or sucking mouth parts and a terminal anus.
- Arthropods possess open circulatory system with heart dorsal in position possessing arteries and no veins, coelom is reduced.
- Respiration occur by gills, tracheae (air ducts), book lungs, or body surface
- Excretion in Arthropods occur by coxal or green glands or by two to many Malpighian tubules on their gut.

- Arthropod nervous system consist of anterior brain (ganglia) connected to paired ventral nerve cords and ganglia with nerves in each body somite or ganglia concentrated anteriorly; sense organs for sight, touch, taste, and smell.
- Sexes are usually separate, fertilization is mostly internal; eggs contains yolk; one or more larval (or nymphal) stages are found; parthenogenesis is common in some crustaceans and insects

	Crustacea (Crustacean)	Insecta (Insects)	Arachnida (Spiders, etc.)	Chilopoda (Centipedes)	Diplopoda (Millipedes)
Body divisions	Usually cephalothorax and abdomen	Head, thorax, abdomen	Cephalothorax and abdomen	Head and long body	Head, short thorax, long abdomen
Antennae	2 pairs	1 pair	None	1 pair	1 pair
Mouth parts	Mandibles, maxillae, 2 pairs maxillipeds	Mandibles Maxillae, 1 pair Labium	Chelicerae Pedipalpi	Mandibles Maxillae, 2 pairs	Mandibles Maxillae, 1 pair
Legs	1 pair per somite, or less	3 pairs on thorax (+ wings)	4 pairs on cephalothorax	1 pair per somite	2 (or 1) pairs per somite
Respire by	Gills or body surface	Tracheae	Book lungs, or tracheae	Tracheae	Tracheae
Sex openings	2, hind part of thorax	1, end of abdomen	1, second somite of abdomen	1, end of abdomen	1, third somite near head
Development	Usually with larval stages	Usually with larval stages	Direct, except mites and ticks	Direct	Direct

PHYLUM - ECHINODERMATA

Body is unsegmented (non metameric) with radial, pentamerous symmetry; body rounded, cylindrical or star-shaped, with five or more radiating areas.

No head or brain; few specialized sensory organs: sensory system of tactile and chemoreceptors, podia, terminal tentacles, photoreceptors and statocysts.

Calcareous endoskeleton in the form of plates or tiny ossicles.

A water-vascular system of coelomic origin that extends from the body surface as a series of tentacle-like projections (podia or tube feet). Coelom is extensive forming the perivisceral cavity and the cavity of the water vascular system

Locomotion by tube feet or by movement of spines or by movement of arms.

Nervous system with radial nerves.

Digestive system usually complete; axial or coiled; anus absent in ophiuroids.

Blood-vascular system (hemal system) much reduced, playing little, if any, role in circulation

Excretory organs absent.

PHYLUM CHORDATA

The phylum Chordata comprises the tunicates, lancelets and the vertebrates – Sharks, bonyfishes, amphibians, reptiles, birds and mammals. The lower chordates are mostly of small size, all are marine, and most tunicates are sessile. The vertebrates include practically all living animals of medium to large size including some huge sharks and whales, Vertebrates occupy all kinds of habitats in salt and fresh waters and on land, most of the reptiles, birds and mammals are strictly terrestrial

GENERAL CHARACTERISTICS OF CHORDATES

- Chordates are bilaterally symmetrical animals with three germ layers.
- They are basically segmented with well developed coelom.
- They possess complete digestive tract – mouth, anus and all other associated organs.

- Three outstanding characteristics distinguishes chordates from all other animals: a single, dorsal tubular nerve cord, a notochord and gill slits in the pharynx.
- The **notochord** is the first supporting structure of the **chordate** body. In the early embryo, it forms above the primitive gut as a slender rod of cells containing gelatinous matrix and is sheathed in fibrous connective tissue. In tunicates, it is present in the tail and only during the larval stages. In higher vertebrates, it is later surrounded or replaced by the vertebral column. The **nerve cord** forms on the dorsal surface of the early embryo soon after the gastrula stage. The anterior end become enlarged as a simple 'cerebral vesicle' in tunicate larvae and in lancelets; but in all vertebrates, it thickens and differentiates as the brain.
- **Paired gill pouches** develop on the sides of the embryonic pharynx. All aquatic chordates from tunicates to amphibians respire by gills. In reptiles, birds and mammals, several gill pouches which develop during early embryonic life are never functional but soon close. All these animals later acquire lungs for breathing air when they hatch or are born.

Characteristics of Subphylum – Vertebrata

- The major diagnostic features of chordates – notochord, dorsal tubular nerve cord, pharyngeal pouches, endostyle or thyroid gland, and postanal tail – all present at some stage of the life cycle.
- Vertebrates possess integument of two divisions: outer epidermis and an inner dermis. Other modifications of the skin include glands, scales, feathers, claws, horns and hair
- They possess distinctive cartilage or bone i.e endoskeleton consisting vertebral column (except In hagfishes which lack vertebrae).
- Vertebrates possess complete, muscularized digestive tract with distinct liver and pancreas.
- Their circulatory system comprises the ventral heart of multiple chambers and closed blood vessel system of arteries, veins and capillaries. Blood contains erythrocytes with haemoglobin.
- They possess well developed coelom divided into a pericardial cavity and pleuroperitoneal cavity
- Vertebrates nearly always possess separate sexes, each sex containing gonads with ducts that discharge their products either into the cloaca or in special openings near the anus.
- Most vertebrates possess two pairs of appendages supported by limb girdles and appendicular skeleton.
- Vertebrates possess highly differentiated tripartite brain with 10 or 12 cranial nerves.

- Vertebrates possess muscular pharynx. In fishes, pharyngeal pouches open to the outside as slits and bear gills

ASSIGNMENT

Outline the major distinguishing features among the different classes of invertebrates and vertebrates using their cellular organization, coelom, body symmetry and reproduction.